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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (MMD) (CBCS) Degree Examinations
 May-June 2025

22MMD202 – FATIGUE OF MATERIALS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Discuss any ten modes of mechanical failure.	10	L*2	1	1
	b) Illustrate Coulomb Mohr's theory for failure of brittle material.	10	L2	1	1
2.	a) Explain, different stages of fatigue failure.	10	L2	1	1
	b) Plot the S-N curve and explain R R Moore rotating beam test.	10	L2	1	1
Unit – II					
3.	a) An unnotched circular rod with a diameter of 10mm is subjected to constant amplitude bending with $S_m=200$ mPa. The material is tempered with $S_u=1240$ mPa. Estimate S_a , S_{max} , S_{min} and R for 50000 cycles. Take surface factor 0.87.	10	L3	2	1
	b) Explain how the microstructure and surface finish influence S-N behavior.	10	L2	2	1
4.	a) Illustrate the true stress-strain behavior.	10	L2	2	1
	b) Explain strain based approach for life estimation.	10	L2	2	1
Unit – III					
5.	a) Illustrate three modes of loading by which a crack can extend.	10	L2	3	1
	b) A very wide SAE 1020 cold rolled thin plate is subjected to constant amplitude uniaxial monotonic loads that produce nominal stress having $S_{max}= 200$ mPa, $S_{min}= -50$ mPa. The material has $S_y=630$ mPa, $S_u=670$ mPa. $E= 207$ GPa and $K_c= 104$ mPa $\sqrt{m}$. Check whether CEFM is applicable or not. Determine Fatigue crack growth rate and Fatigue life for an initial through thickness edge crack of 1 mm in length. Take $A=6.9 \times 10^{-12}$, $n=3$ and, ≤ 1.12 .	10	L3	3	1
6.	a) Illustrate the notch stress and strain determination by Neuber's rule.	10	L2	3	1
	b) Describe Glinka's rule for analyzing stress and strain at the notch.	10	L2	3	1
Unit – IV					
7.	a) Explain Palmgren-Miner linear damage rule.	10	L2	4	1
	b) Explain racetrack counting and level crossing method for cycle counting.	10	L2	4	1
8.	a) Illustrate two parameter Weibull distribution.	10	L2	4	1
	b) Discuss the following i) Stochastic variable ii) Sample iii) Median iv) Standard deviation v) Variance.	10	L2	4	1
Unit – V					
9.	a) Briefly explain the factors that affect adhesion.	10	L2	5	1
	b) Discuss various precautions that a designer may take to reduce the chances of a wear failure.	10	L2	5	1
10.	a) Explain the three basic weldment regions.	10	L2	5	1
	b) Explain ways of improving weldment fatigue resistance.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
